Chapter 1: Introduction to Engineering Economics

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1. Technical Efficiency

Formula:

$$\boxed{\text{Technical Efficiency} = \frac{\text{Output}}{\text{Input}}}$$

Variable Definitions:

Symbol	Full Form	Unit	Description
Output	Physical Output	units, kW, hp	Useful work produced
Input	Physical Input	fuel, raw material	Resources consumed

When to Use:

Analyzing physical systems (diesel engines, machines, furnaces)
Measuring equipment performance
Comparing efficiency of different machines

Example Problem:

Statement: *"A diesel engine consumes 5 liters of fuel per hour and produces 100 kW of power. Calculate its technical efficiency if 1 liter fuel = 25 kW potential."*

Solution:

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Input (potential) = 5 × 25 = 125 kW
Output (actual) = 100 kW
Technical Efficiency = 100/125 = 0.80 = 80%
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2. Economic Efficiency (Productivity)

Formula:

$$\boxed{\text{Economic Efficiency} = \frac{\text{Worth}}{\text{Cost}} = \frac{\text{Annual Revenue}}{\text{Annual Expenses}}}$$

Variable Definitions:

Symbol	Full Form	Unit	Description
Worth	Annual Revenue	Rs.//$year|Total income generated|Cost|Annual Expenses|Rs.//$year	Total costs incurred|

When to Use:

Evaluating overall business performance
Comparing economic viability of alternatives
Measuring organizational productivity
Investment decision making

Example Problem:

Statement: *"A manufacturing business generates Rs. 50 lakhs annual revenue with Rs. 40 lakhs annual expenses. Calculate economic efficiency."*

Solution:

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Economic Efficiency = 50/40 = 1.25
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Interpretation: For every Re. 1 spent, business generates Rs. 1.25 (25% profit margin)

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3. Five Methods to Improve Economic Efficiency

Method 1 : Increased Output for Same Input

Strategy: Keep input constant → Increase output

Formula:

$$\boxed{\text{New Efficiency} = \frac{\text{Output}_{\text{new}} \uparrow}{\text{Input}_{\text{same}}} > \frac{\text{Output}_{\text{old}}}{\text{Input}_{\text{same}}}}$$

When to Use: When you have spare capacity or can improve worker productivity

Example: Improve worker training → same workers produce more units

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Method 2 : Decreased Input for Same Output

Strategy: Keep output constant → Reduce input costs

Formula:

$$\boxed{\text{New Efficiency} = \frac{\text{Output}_{\text{same}}}{\text{Input}_{\text{new}} \downarrow} > \frac{\text{Output}_{\text{same}}}{\text{Input}_{\text{old}}}}$$

When to Use: When you can reduce waste, find cheaper suppliers, or optimize processes

Example: Reduce material waste → same output with less raw material

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Method 3 : Proportionate Increase Strategy

Concept: Increase both, but revenue ↑ > cost ↑

Formula:

$$\boxed{\frac{\Delta\text{Revenue}}{\text{Revenue}_{\text{old}}} > \frac{\Delta\text{Cost}}{\text{Cost}_{\text{old}}}}$$

Where: | Symbol | Full Form | Description | |——|———|————-| | Δ | Delta (Change) | Difference between new and old |

Example Problem:

Statement: *"A company has idle machinery. It introduces a new product. Revenue increases 30% while costs increase only 15%. Show productivity improves."*

Solution:

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% Revenue increase = 30%
% Cost increase = 15%
Since 30% > 15%, proportionate revenue increase > cost increase
 Economic Efficiency improves
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Method 4 : Proportionate Decrease Strategy

Concept: Decrease both, but cost ↓ > revenue ↓

Formula:

$$\boxed{\frac{\Delta \text{Cost}}{\text{Cost}_{\text{old}}} > \frac{\Delta \text{Revenue}}{\text{Revenue}_{\text{old}}}}$$

Example Problem:

Statement: *"A company drops an unprofitable product. Revenue decreases 10% but costs decrease 25%. Analyze the impact."*

Solution:

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% Revenue decrease = 10%
% Cost decrease = 25%
Since 25% > 10%, proportionate cost decrease > revenue decrease
 Economic Efficiency improves
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Method 5 : Simultaneous Increase-Decrease Strategy

Concept: Output ↑ while Input ↓

Formula:

$$\boxed{\text{New Efficiency} = \frac{\text{Output}_{\text{increased}} \uparrow}{\text{Input}_{\text{decreased}} \downarrow} \gg \text{Old Efficiency}}$$

Where: | Abbreviation | Full Form | |————–|———–| | AGVS | Automated Guided Vehicle System | | O&M | Operation and Maintenance |

Example Problem:

Statement: *"A factory installs robots. Production increases 40% while labor costs decrease 30%. Calculate efficiency improvement."*

Solution:

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Let old efficiency = Output/Input = 100/100 = 1.0
New output = 140 (40% increase)
New input = 70 (30% decrease)
New efficiency = 140/70 = 2.0
Improvement = (2.0 - 1.0)/1.0 = 100% improvement
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4. Marginal Cost (MC)

Formula:

$$\boxed{MC = \frac{\Delta TC}{\Delta Q} = \frac{TC_2 - TC_1}{Q_2 - Q_1}}$$

Variable Definitions:

Symbol	Full Form	Unit	Description
MC	Marginal Cost	Rs./unit, $/unit	Cost of one additional unit
ΔTC	Change in Total Cost	Rs., $	TC - TC
ΔQ	Change in Quantity	units	Q - Q
TC	Total Cost	Rs., $	All costs combined
Q	Quantity/Output	units	Production volume

When to Use:

Finding cost of producing ONE MORE unit
Production optimization decisions
Determining if additional production is profitable
Profit maximization (optimal when MC = MR)

Example Problem:

Statement: *"Total cost increases from Rs. 1000 to Rs. 1200 when output increases from 10 to 12 units. Find marginal cost."*

Given: - TC = Rs. 1000, Q = 10 units - TC = Rs. 1200, Q = 12 units

Solution:

$$MC = \frac{1200 - 1000}{12 - 10} = \frac{200}{2} = \boxed{\text{Rs. 100 per unit}}$$

Interpretation: Each additional unit costs Rs. 100 to produce

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5. Marginal Revenue (MR)

Formula:

$$\boxed{MR = \frac{\Delta TR}{\Delta Q} = \frac{TR_2 - TR_1}{Q_2 - Q_1}}$$

Variable Definitions:

Symbol	Full Form	Unit	Description
MR	Marginal Revenue	Rs./unit, $/unit	Revenue from one additional unit
ΔTR	Change in Total Revenue	Rs., $	TR - TR
ΔQ	Change in Quantity	units	Q - Q
TR	Total Revenue	Rs., $	Total sales income
Q	Quantity Sold	units	Sales volume

When to Use:

Finding revenue from selling ONE MORE unit
Pricing decisions
Profit maximization (optimal when MR = MC)
Analyzing impact of sales volume changes

Example Problem:

Statement: *"Total revenue increases from Rs. 5000 to Rs. 5400 when sales increase from 100 to 108 units. Find marginal revenue."*

Given: - TR = Rs. 5000, Q = 100 units - TR = Rs. 5400, Q = 108 units

Solution:

$$MR = \frac{5400 - 5000}{108 - 100} = \frac{400}{8} = \boxed{\text{Rs. 50 per unit}}$$

Interpretation: Each additional unit sold generates Rs. 50 revenue

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6. Sunk Cost

Definition:

Sunk Cost = Past cost of equipment/asset that **CANNOT** be recovered

Key Rule:

$$\boxed{\text{Sunk Cost} \Rightarrow \text{IGNORE in future decisions}}$$

Use: Current market value/salvage value
Don't use: Original purchase price (historical cost)

When to Use This Concept:

Replacement analysis
Equipment upgrade decisions
Any future investment decision

Example Problem:

Statement: *"Equipment was purchased for Rs. 1,00,000 three years ago. Its current market value is Rs. 60,000. What value should be used for replacement analysis?"*

Item	Amount	Use in Analysis?	Reason
Original purchase price	Rs. 1,00,000	NO	Sunk cost (already spent)
Current market value	Rs. 60,000	YES	Opportunity cost if kept

Answer: Rs. 60,000 (current market value)

Why? The Rs. 1,00,000 is already spent and cannot be recovered. Only the current Rs. 60,000 value matters for future decisions.

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7. Opportunity Cost

Definition:

Opportunity Cost = Value of the **best alternative foregone** (given up)

Formula (Implicit):

$$\boxed{\text{Opportunity Cost of A} = \text{Benefit of Best Alternative B} - \text{Benefit of A}}$$

When to Use:

- Comparing mutually exclusive alternatives
- Resource allocation decisions
- Capital budgeting
- Government policy choices

Decision Rule: Choose option with **MINIMUM** opportunity cost

Example Problem:

Statement: *"Government has Rs. 10 billion budget. Option A: Invest in National Health Service (benefits 5 million people). Option B: Invest in Education (benefits 8 million people). If government chooses A, what is the opportunity cost?"*

Analysis:

Option	Investment	Direct Benefit
A: Health Service	Rs. 10 billion	5 million people benefited
B: Education (foregone)	Rs. 10 billion	8 million people benefited

Opportunity Cost of choosing A:

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= Benefit of Education (Best Alternative)
= Educational benefits for 8 million people
= Rs. 10 billion worth of educational improvement
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Answer: The opportunity cost is the educational benefits foregone (worth Rs. 10 billion)

Key Insight: Every choice has a cost = value of next best alternative you sacrifice

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Summary Table: Key Formulas Chapter 1

Formula	Variables	When to Use	Example Context
Technical Eff. = Output/Input	Output, Input	Physical system analysis	Engine efficiency
Economic Eff. = Revenue/Cost	Revenue, Cost	Business performance	Profitability analysis
MC = ΔTC/ΔQ	TC, Q	Production decisions	Cost of next unit
MR = ΔTR/ΔQ	TR, Q	Pricing decisions	Revenue from next sale
Ignore Sunk Cost	-	Replacement analysis	Equipment decisions
Opportunity Cost	Benefits of alternatives	Choice between options	Resource allocation